

Instruction manual

suspension bridge



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MARS EDPAL INSTRUMENT PVT LTD

SUSPENSION BRIDGE APPARATUS

INSTRUCTION MANUAL

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1.0 APPARATUS REQUIRED:

- Portal frame Apparatus
- Digital Balance : 2 Nos.

2.0 OBJECTIVE:

- Comparison of simplified theory with experimental results for a uniformly distributed load
- Study of the effect of a point load

3.0 THEORY:

The suspension bridge is defined as a bridge that has a roadway supported by cables that are anchored at both ends.

Essentially, a suspension bridge consists of the following elements.

- i. The cable,
- ii. Decking.
- iii. Supporting tower

The traffic load of the decking is transferred to main cable through the suspenders. Since the cable is the main load bearing member, the curvature of the cable of an unstiffened bridge change as the load moves on the decking. To avoid this, the decking is stiffened by provision of either a three hinged stiffened girder. The stiffened girder transfers a uniformly distributed or equal load to each suspender, irrespective of the load position on the decking. The suspension cable is supported on either side. There are two arrangements generally used. The suspension cable may either pass over a smooth frictionless pulley and anchored to the other side, or it may be attached to a saddle placed on rollers. In the former case the tension in the cable on the two sides of the pulley are equal while in the latter case, the horizontal components of the tension on the two sides are equal since the cable cannot have a movement relative to the saddle. The cable consists of the either wire, rope, parallel wires jointed with clips or eye bar links. The cable can carry direct tension only, and the bending moment at any point on the cable is zero. The suspenders consist of round rods or ropes with turn buckler so that adjustment in their lengths may be done if required.

The anchorage consists of huge mass of concrete, designed to resist the tension of the cable. Let us take the case of a cable geometry having the cable chord horizontal.

4.0 DESCRIPTION:

A rigid bridge deck is carried each side by hangers of different lengths such that the flexible steel cables from which they hang are constrained to a parabolic curve. The twin suspension cables pass over pulleys each end of the 1 m span and terminate in yokes carrying load hangers. Adjustable stops prevent the yokes moving upward and mark the correct length from the cables. Steel bars are provided to simulate a uniformly distributed load, and a special point load is supplied.

5.0 PROCEDURE:

1. Set the spring balance on zero positions
2. Apply 1 kg load at each hanger (any two) hangers for remove any play or slackness & Note down the initial reading.
3. Then apply equal load (any) left and right and note down the final reading of the spring balance.
4. Measure the readings with the help of spring balance.
5. Note down the all reading and take the average reading 'T' tension practically.
6. Experimental Value of 'T' is determined & Compare practically & experimentally.

6.0 OBSERVATION & CALCULATIONS:

Cable Length = foot (Approx)

T = Cable Tension

d = Diameter of Cable = 3 mm

A = Area of Cross-section (Cable) = $\pi d^2/4 = 7.06 \text{ mm}^2$

P = Point Load

L = Span Length = 78 cm = 0.78 m

Spring balance reading show the distributed load (W)

W_1 = Left side balance readings.

W_2 = Right side balance readings.

σ = Stress on the cable

$$T_{\text{theory}} = W/2$$

$$\sigma = F/A = (T/2A)$$

According to the data in the lab, fill the following table:

S.No	Point Load (P)	T_1 (W_1)/2	T_2 (W_2)/2	Tension (T_{exp}) ($T_1 + T_2$)/2	Tension (T_{theory}) (P/2)	Stress (σ) (T/2A)
1.	2 kg	0.6/2	0.5	0.55	0.5	
2.	2 kg	1/2	1.3/2	1.13/2	2/2	

S.No	Point Load (P) in N	Tension (T _{exp}) in N	Tension (T _{theory}) in N	Stress (N/mm ²)
1.	9.81 N	11.57 N	11.57 N	
2.				

Draw the relation with the load (N) with the experimental value of the tension (N).

7.0 RESULT:

Experimentally find out the cable tension (T):. _____